Adaptive Staircase Experiment
PSY310 Lab in Psychology
Lab Report

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Tamanna Chandna
AU2420048

GitHub Link: https://github.com/tamannaz06/Staircase-method

Introduction

Adaptive staircase method is an algorithm that actively adjusts the stimuli on-line in response to the subject's performance (Hairston & Joseph, 2008). Meaning, it adjusts becoming easier or harder depending on how the participants respond. The procedure adjusts according to pre-set rules and terminates when the stimuli approximate to the subject's threshold, making it a much more efficient method at the cost of lower data availability (Hairston & Joseph, 2008). This method is used in psychophysics very often to determine the participants' sensory thresholds. This method of testing was first statistically studied and analyzed by Dixon and Mood (1948).

It is essentially a variation of the method of limits in which stimuli are presented in ascending and descending order. When the observer's response changes, the direction of the stimulus sequence is reversed. This method is efficient because it does not present stimuli that are well above or below threshold. (APA Dictionary)

In this current study, the adaptive staircase procedure is used to estimate the difference threshold or just noticeable difference (JND) for knowing the orientation discrimination. The JND essentially represents the smallest tilt change that would be required to detect a change. In this manner, it measures the participant's perceptual sensitivity.

Some real life examples that have used this method for its studies are explosives testing and later for fatigue testing. (Little, 1975a)

Method

Participant

This experiment has one participant who is a 20-year-old male undergraduate 3rd year student of Ahmedabad University. Informed consent was taken from the participant and his data was kept anonymous by assigning the experiment name a random number generated by PsychoPy. He was given clear instructions which was to determine the direction of the tilt of the stimuli shown on the laptop display.

Materials and stimuli

The experiment was conducted on a laptop with PsychoPy Studio installed. The laptop has a monitor of dimension 13.3", a resolution of 1920x1080 pixels and a refresh rate of 60 Hz. 100 trials of the experiment were conducted.

The stimulus was a sinusoidal shape with a Gaussian mask. The following figure shows this grating in a left and right tilt.

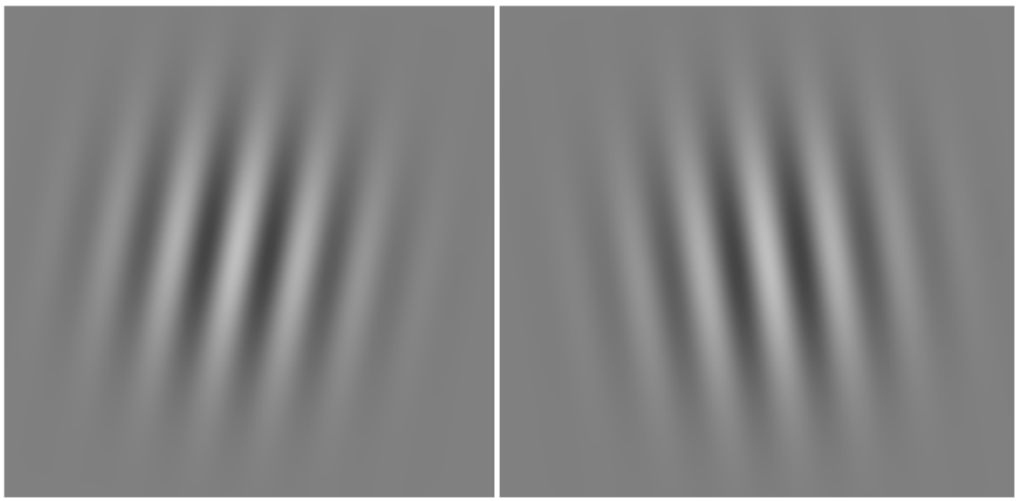


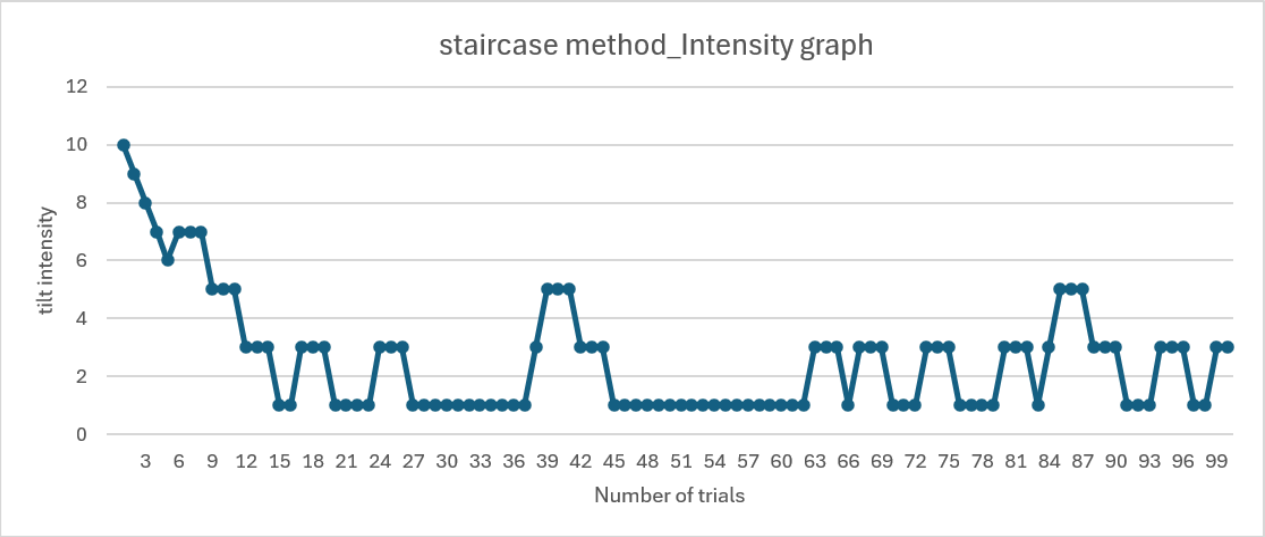
Figure: Stimuli in left and right grating.

Procedure

The experiment started with the instruction page. There was no practice trial but it was ensured that the instructions were clear. A sample of the gratings were shown.

Participants were instructed to indicate whether the stimuli were tilted to the right or left using the arrow keys [→] and [←] respectively. Each trial began with a cross-shaped fixation shown for 500 milliseconds. After this, the stimuli (sinusoidal shape with a gaussian mask) appeared for 300 milliseconds. The tilt of the grating on each trial was either left or right-tilted, based on random probability. This routine of the fixation and grating continued for 100 trials. The tilt intensity was adjusted using the step sizes of 2, 1, 1 and 0.5, the routine followed a one-up, three-down adaptive staircase procedure. The maximum tilt was 10 and minimum 1.

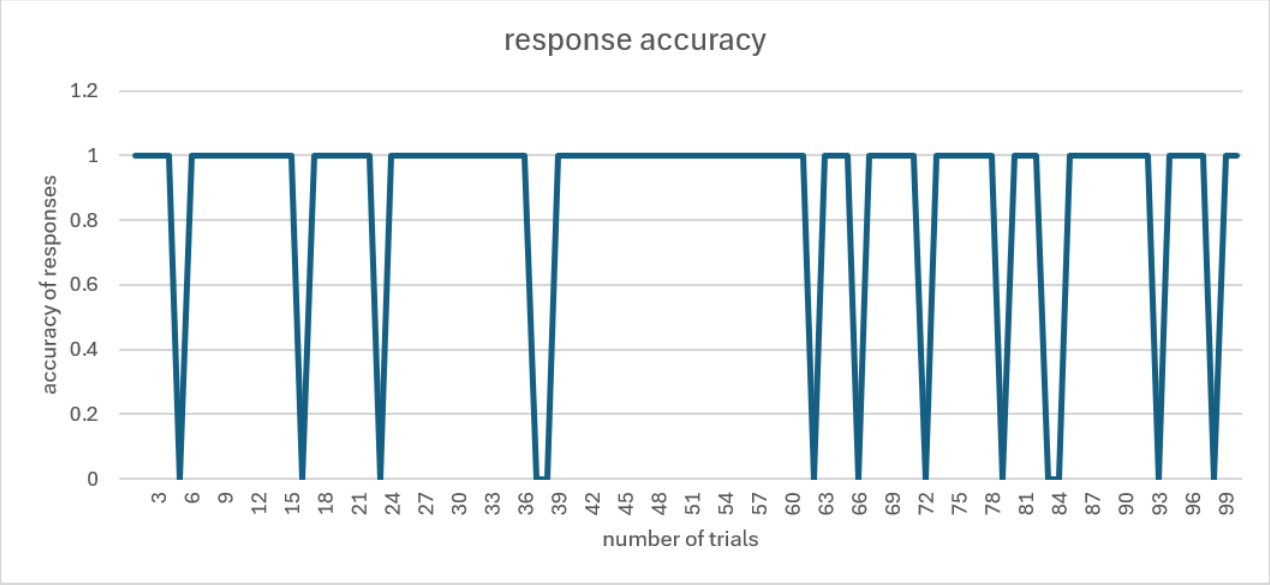
The tilt intensity across the 100 trials is plotted on the graph in Figure 2. Responses were collected using a standard keyboard via PsychoPy and saved in an Excel file. This data was further used to calculate the threshold and accuracy.



A graph representing the changes in tilt intensity across the extent of 100 trials of the procedure

Results

The absolute difference threshold is 1, after which the participant could no longer identify the tilt. This was calculated by first identifying the last five 1-to-0 reversals in the data-set of 100 trials and then calculating the average of those intensities. The participant had an accuracy rate of 87% across the 100 trials, gradually getting better as the trials progressed. This is quite a high accuracy. The figure below shows the plotted accuracy for each trial across the 100.



Graph representing the accuracy of responses over the 100 trials. (The x axis has intervals of 3 for ease of representation).

Discussion

The difference threshold of 1 degree is the point at which the participant could no longer reliably discriminate between left- and right-tilted stimuli. It's also the lowest tilt that was measured so the participant had a really low JND which is indicative of very high perceptual sensitivity; the participant required small differences to make judgements that were accurate.
Further, The accuracy rate of 87% suggests that the participant performed above much above chance slowly having even more accurate answers.
In optimal conditions with more trials, the participant is more likely to be influenced by practice effects, making their response accuracy more precise towards the latter half of the trials. It is also worth noting that the validity of this threshold in external environments beyond the lab environment where this was conducted is unreliable. So, due to the presence of various possibilities of confounding stimuli, one would need to replicate this experiment under different conditions to gain external validity. This is why further research is required.

References

1) American Psychological Association. (n.d.). Staircase method. In *APA dictionary of psychology*. https://dictionary.apa.org/staircase-method

2) Hairston, D., & Joseph, M. (2008, October 5). An adaptive staircase procedure for the E-prime programming environment - sciencedirect. https://www.sciencedirect.com/science/article/abs/pii/S016926070800206X

3) R, P., Palazotto, A., & Nicholas, T. (2006, July 7). A simulation-based investigation of the staircase method for fatigue strength testing - sciencedirect. https://www.sciencedirect.com/science/article/abs/pii/S0167663606000184

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